



AMERICAN TOWER™

Análisis de Torre por Viento

Estructura : 42.0 m MONOPOLO

Nombre Sitio CFE : CAPULHUAC

ID CFE : 9151016916R

Ingeniería Número : 179027-100003

Carrier : CFE TEIT

Solicitud (folio) : COLO SIT-12706003

País : México

Fecha : May 30, 2024

Eficiencia General : 67%

Emitido por:

Departamento de Ingeniería
Shared Service Center Operations LatAm

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

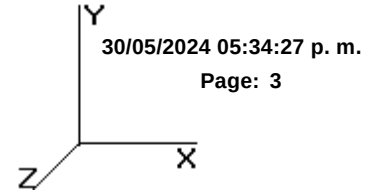
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Shaft Section Properties					Slip		Bottom						Top							
Sect	Length	Thick	Fy	Joint	Join	Weight	Dia	Elev	Area	Ix	W/t	D/t	Dia	Elev	Area	Ix	W/t	D/t	Taper	
Info	(m)	(cm)	(kg/ cm²)	Type	Len	(ton)	(cm)	(m)	(cm²)	(cm^4)	Ratio	Ratio	(cm)	(m)	(cm²)	(cm^4)	Ratio	Ratio	(cm/m)	
1-R	6.10	0.9750	2530		0.00	1.551	106.700	0.00	323.84	452824.8	0.00	109.44	106.700	6.10	323.84	452824.8	0.00	109.44	0.000000	
2-R	5.60	1.0030	2530	Butt	0.00	1.252	91.400	6.10	284.84	291176.0	0.00	91.13	91.400	11.70	284.84	291176.0	0.00	91.13	0.000000	
3-R	6.10	0.9800	2530	Butt	0.00	1.109	76.200	11.70	231.58	163914.9	0.00	77.76	76.200	17.80	231.58	163914.9	0.00	77.76	0.000000	
4-R	6.10	0.6970	2530	Butt	0.00	0.792	76.200	17.80	165.33	117901.1	0.00	109.33	76.200	23.90	165.33	117901.1	0.00	109.33	0.000000	
5-R	6.10	0.6970	2530	Butt	0.00	0.792	76.200	23.90	165.33	117901.1	0.00	109.33	76.200	30.00	165.33	117901.1	0.00	109.33	0.000000	
6-R	6.10	0.6970	2530	Butt	0.00	0.632	61.000	30.00	132.04	60067.9	0.00	87.52	61.000	36.10	132.04	60067.9	0.00	87.52	0.000000	
7-R	6.10	0.7560	2530	Butt	0.00	0.685	61.000	36.10	143.08	64961.5	0.00	80.69	61.000	42.20	143.08	64961.5	0.00	80.69	0.000000	
Shaft Weight						6.812														

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
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Base Elev : 0.500 (m)



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Discrete Appurtenance Properties

Attach Elev (m)	Description	Qty	No Ice			Ice			Distance From Face (m)	Vert Ecc (m)
			Weight (ton)	CaAa (m ²)	CaAa Factor	Weight (ton)	CaAa (m ²)	CaAa Factor		
42.00	Lightning Rod	1	0.03	0.093	1.00	0.00	0.000	1.00	0.000	0.914
40.00	DISH HP Ø1290	1	0.09	1.650	0.89	0.00	0.000	1.00	0.000	0.000
37.20	PANEL 2769x469x206	3	0.03	1.755	0.76	0.00	0.000	1.00	0.000	0.000
37.20	RRU 488x290x168	6	0.03	0.169	0.79	0.00	0.000	1.00	0.000	0.000
37.20	T-Arm	3	0.06	0.800	0.75	0.00	0.000	1.00	0.000	0.000
32.20	DISH HP Ø667	1	0.04	0.441	0.89	0.00	0.000	1.00	0.000	0.000
32.20	Side Arm	1	0.02	0.470	1.00	0.00	0.000	1.00	0.000	0.000
30.60	GPS	1	0.03	0.016	1.00	0.00	0.000	1.00	0.000	0.000
29.40	BBU	1	0.03	0.182	1.00	0.00	0.000	1.00	0.000	0.000
29.40	Power	1	0.03	0.144	1.00	0.00	0.000	1.00	0.000	0.000
29.00	PANEL 2535x298x149	3	0.03	1.095	0.80	0.00	0.000	1.00	0.000	0.000
29.00	RRU 400x300x150	3	0.03	0.144	0.75	0.00	0.000	1.00	0.000	0.000
29.00	Side Arm	3	0.02	0.470	0.67	0.00	0.000	1.00	0.000	0.000
28.60	OTRO 600x240x430	1	0.03	0.310	0.78	0.00	0.000	1.00	0.000	0.000
Totals		29	0.98			0.00			Number of Loadings : 14	

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Linear Appurtenance Properties Load Case Azimuth (°) :

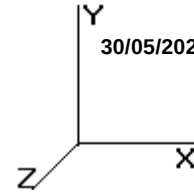
Elev To (m)	Elev From (m)	Description	Coax Dia (cm)	Exposed Width (m)	Exposed To Wind	Coax Wt (kg/m)	Flat	Max Coax / Row	Dist Between Rows	Dist Between Cols (cm)	Azimuth (deg)	Dist From Face (cm)	Carrier
0.00	42.20	(1) CGO	3.81	0.04	Y	8.93	Y	1	0.00	0.00	90	30.00	
0.00	30.00	(1) CGO	3.81	0.04	Y	8.93	Y	1	0.00	0.00	270	30.00	CGO
0.00	42.20	(1) Climbing Ladder, Int	4.76	0.05	Y	10.27	Y	1	0.00	0.00	0	30.00	
0.00	37.00	(1) Coax 4.28 cm	4.28	0.04	Y	2.80	N	1	0.00	0.00	90	32.00	AT&T
0.00	30.00	(1) Coax 4.74 cm	4.74	0.05	Y	1.00	N	1	0.00	0.00	270	32.00	CFE-

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Segment Properties

(Max Len : 1.50 (m))

Seg							F'y	S	Z		Additional Reinforcing		
Top	Thick	Flat Dia	Area	Ix	W/t Ratio	D/t Ratio	(kg/	(cm ²)	(cm ³)	Weight	Area	Ix	Weight
Elev (m)	(cm)	(cm)	(cm ²)	(cm ⁴)			cm ²)			(ton)	(cm ²)	(cm ⁴)	(ton)
0.00	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.00	FixAr	FixI	FixWt
1.50	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.38	FixAr	FixI	FixWt
3.00	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.38	FixAr	FixI	FixWt
4.50	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.38	FixAr	FixI	FixWt
6.00	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.38	FixAr	FixI	FixWt
6.10	0.9750	106.70	323.841	452824.8	0.00	109.44	2392.8	8487.8	10898.6	0.03	FixAr	FixI	FixWt
6.10	1.0030	91.400	284.843	291176.0	0.00	91.13	2530.0	6371.5	8196.5		FixAr	FixI	FixWt
7.50	1.0030	91.400	284.843	291176.0	0.00	91.13	2530.0	6371.5	8196.5	0.31	FixAr	FixI	FixWt
9.00	1.0030	91.400	284.843	291176.0	0.00	91.13	2530.0	6371.5	8196.5	0.34	FixAr	FixI	FixWt
10.50	1.0030	91.400	284.843	291176.0	0.00	91.13	2530.0	6371.5	8196.5	0.34	FixAr	FixI	FixWt
11.70	1.0030	91.400	284.843	291176.0	0.00	91.13	2530.0	6371.5	8196.5	0.27	FixAr	FixI	FixWt
11.70	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2		FixAr	FixI	FixWt
12.00	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2	0.05	FixAr	FixI	FixWt
13.50	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2	0.27	FixAr	FixI	FixWt
15.00	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2	0.27	FixAr	FixI	FixWt
16.50	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2	0.27	FixAr	FixI	FixWt
17.80	0.9800	76.200	231.584	163914.9	0.00	77.76	2530.0	4302.2	5545.2	0.24	FixAr	FixI	FixWt
17.80	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5		FixAr	FixI	FixWt
18.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.03	FixAr	FixI	FixWt
19.50	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
21.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
22.50	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
23.90	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.18	FixAr	FixI	FixWt
23.90	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5		FixAr	FixI	FixWt
24.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.01	FixAr	FixI	FixWt
25.50	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
27.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
28.50	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.19	FixAr	FixI	FixWt
28.60	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.01	FixAr	FixI	FixWt
29.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.05	FixAr	FixI	FixWt
29.40	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.05	FixAr	FixI	FixWt
30.00	0.6970	76.200	165.328	117901.1	0.00	109.33	2393.5	3094.5	3973.5	0.08	FixAr	FixI	FixWt
30.00	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7		FixAr	FixI	FixWt
30.60	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.06	FixAr	FixI	FixWt
31.50	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.09	FixAr	FixI	FixWt
32.20	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.07	FixAr	FixI	FixWt
33.00	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.08	FixAr	FixI	FixWt
34.50	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.16	FixAr	FixI	FixWt
36.00	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.16	FixAr	FixI	FixWt
36.10	0.6970	61.000	132.045	60067.9	0.00	87.52	2530.0	1969.4	2534.7	0.01	FixAr	FixI	FixWt
36.10	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9		FixAr	FixI	FixWt
37.20	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.12	FixAr	FixI	FixWt
37.50	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.03	FixAr	FixI	FixWt
39.00	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.17	FixAr	FixI	FixWt
40.00	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.11	FixAr	FixI	FixWt
40.50	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.06	FixAr	FixI	FixWt
42.00	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.17	FixAr	FixI	FixWt
42.20	0.7560	61.000	143.082	64961.5	0.00	80.69	2530.0	2129.9	2743.9	0.02	FixAr	FixI	FixWt

6.8

Pole : 179027
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Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
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Load Case: 1.0D + 1.0W

96.56 km/hrServiceability

26 Iterations

Gust Response Factor :1.10
Dead Load Factor :1.00
Wind Load Factor :1.00

Wind Importance Factor 0.00

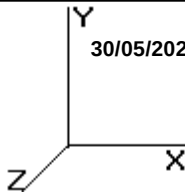
Applied Segment Forces Summary

Seg Elev (m)	Lateral FX (-) (ton)	Axial FY (-) (ton)	Torsion MY (ton- m)	Moment MZ (ton- m)
0.00	0.01	0.00	0.00	0.00
1.50	0.04	0.43	0.00	0.00
3.00	0.04	0.43	0.00	0.00
4.50	0.04	0.43	0.00	0.00
6.00	0.03	0.43	0.00	0.00
6.10	0.01	0.03	0.00	0.00
7.50	0.04	0.36	0.00	0.00
9.00	0.05	0.38	0.00	0.00
10.50	0.04	0.38	0.00	0.00
11.70	0.03	0.31	0.00	0.00
12.00	0.02	0.06	0.00	0.00
13.50	0.04	0.32	0.00	0.00
15.00	0.05	0.32	0.00	0.00
16.50	0.04	0.32	0.00	0.00
17.80	0.03	0.28	0.00	0.00
18.00	0.02	0.03	0.00	0.00
19.50	0.05	0.24	0.00	0.00
21.00	0.05	0.24	0.00	0.00
22.50	0.05	0.24	0.00	0.00
23.90	0.03	0.23	0.00	0.00
24.00	0.02	0.02	0.00	0.00
25.50	0.05	0.24	0.00	0.00
27.00	0.05	0.24	0.00	0.00
28.50	0.04	0.24	0.00	0.00
28.60	0.01	0.05	0.00	0.00
29.00	0.14	0.30	0.00	0.00
29.40	0.03	0.12	0.00	0.00
30.00	0.02	0.10	0.00	0.00
30.60	0.02	0.11	0.00	0.00
31.50	0.02	0.11	0.00	0.00
32.20	0.06	0.16	0.00	0.00
33.00	0.03	0.10	0.00	0.00
34.50	0.04	0.19	0.00	0.00
36.00	0.03	0.19	0.00	0.00
36.10	0.01	0.01	0.00	0.00
37.20	0.21	0.58	0.00	0.00
37.50	0.02	0.04	0.00	0.00
39.00	0.03	0.20	0.00	0.00
40.00	0.09	0.22	0.00	0.00
40.50	0.02	0.07	0.00	0.00
42.00	0.03	0.23	0.00	0.00
42.20	0.00	0.03	0.00	0.00
Totals:	1.70	9.00	0.00	0.00

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Load Case: 1.0D + 1.0W

96.56 km/hr Serviceability

26 Iterations

Gust Response Factor :1.10
Dead Load Factor :1.00
Wind Load Factor :1.00

Wind Importance Factor 0.00

Calculated Forces

Seg Elev (m)	Pu FY (-) (ton)	Vu FX (-) (ton)	Tu MY (ton-m)	Mu MZ (ton-m)	Mu MZ (ton-m)	Resultant Moment (ton-m)	phi Pn (ton)	phi Vn (ton)	phi Tn (ton-m)	phi Mn (ton-m)	Total Deflect (cm)	Rotation (deg)	Ratio
0.00	-9.00	-1.69	0.00	-42.61	0.00	42.61	697.39	221.22	246.74	222.72	0.00	0.00	20.427
1.50	-8.57	-1.65	0.00	-40.08	0.00	40.08	697.39	221.22	246.74	222.72	0.05	-0.04	19.229
3.00	-8.14	-1.61	0.00	-37.60	0.00	37.60	697.39	221.22	246.74	222.72	0.20	-0.07	18.055
4.50	-7.71	-1.57	0.00	-35.18	0.00	35.18	697.39	221.22	246.74	222.72	0.44	-0.11	16.908
6.00	-7.28	-1.54	0.00	-32.83	0.00	32.83	697.39	221.22	246.74	222.72	0.77	-0.14	15.789
6.10	-7.25	-1.53	0.00	-32.67	0.00	32.67	697.39	221.22	246.74	222.72	0.79	-0.14	15.716
6.10	-7.25	-1.53	0.00	-32.67	0.00	32.67	648.59	194.58	185.56	171.63	0.79	-0.14	20.162
7.50	-6.90	-1.49	0.00	-30.54	0.00	30.54	648.59	194.58	185.56	171.63	1.18	-0.17	18.861
9.00	-6.51	-1.45	0.00	-28.30	0.00	28.30	648.59	194.58	185.56	171.63	1.68	-0.21	17.501
10.50	-6.13	-1.40	0.00	-26.14	0.00	26.14	648.59	194.58	185.56	171.63	2.29	-0.25	16.178
11.70	-5.82	-1.37	0.00	-24.45	0.00	24.45	648.59	194.58	185.56	171.63	2.84	-0.28	15.149
11.70	-5.82	-1.37	0.00	-24.45	0.00	24.45	527.32	158.20	125.54	118.98	2.84	-0.28	21.664
12.00	-5.76	-1.36	0.00	-24.04	0.00	24.04	527.32	158.20	125.54	118.98	2.99	-0.29	21.305
13.50	-5.43	-1.32	0.00	-22.00	0.00	22.00	527.32	158.20	125.54	118.98	3.83	-0.35	19.530
15.00	-5.11	-1.27	0.00	-20.02	0.00	20.02	527.32	158.20	125.54	118.98	4.81	-0.40	17.807
16.50	-4.79	-1.23	0.00	-18.11	0.00	18.11	527.32	158.20	125.54	118.98	5.92	-0.45	16.139
17.80	-4.51	-1.20	0.00	-16.51	0.00	16.51	527.32	158.20	125.54	118.98	6.99	-0.49	14.741
17.80	-4.51	-1.20	0.00	-16.51	0.00	16.51	356.14	112.94	89.96	81.21	6.99	-0.49	21.612
18.00	-4.48	-1.18	0.00	-16.27	0.00	16.27	356.14	112.94	89.96	81.21	7.16	-0.49	21.307
19.50	-4.24	-1.14	0.00	-14.50	0.00	14.50	356.14	112.94	89.96	81.21	8.53	-0.55	19.053
21.00	-3.99	-1.09	0.00	-12.79	0.00	12.79	356.14	112.94	89.96	81.21	10.04	-0.60	16.885
22.50	-3.75	-1.04	0.00	-11.16	0.00	11.16	356.14	112.94	89.96	81.21	11.66	-0.64	14.806
23.90	-3.53	-1.00	0.00	-9.71	0.00	9.71	356.14	112.94	89.96	81.21	13.27	-0.68	12.950
23.90	-3.53	-1.00	0.00	-9.71	0.00	9.71	356.14	112.94	89.96	81.21	13.27	-0.68	12.950
24.00	-3.51	-0.99	0.00	-9.61	0.00	9.61	356.14	112.94	89.96	81.21	13.39	-0.68	12.822
25.50	-3.27	-0.93	0.00	-8.13	0.00	8.13	356.14	112.94	89.96	81.21	15.21	-0.71	10.930
27.00	-3.02	-0.88	0.00	-6.72	0.00	6.72	356.14	112.94	89.96	81.21	17.10	-0.74	9.134
28.50	-2.78	-0.84	0.00	-5.40	0.00	5.40	356.14	112.94	89.96	81.21	19.06	-0.76	7.438
28.60	-2.74	-0.83	0.00	-5.32	0.00	5.32	356.14	112.94	89.96	81.21	19.20	-0.76	7.321
29.00	-2.44	-0.69	0.00	-4.99	0.00	4.99	356.14	112.94	89.96	81.21	19.73	-0.76	6.828
29.40	-2.31	-0.66	0.00	-4.71	0.00	4.71	356.14	112.94	89.96	81.21	20.26	-0.77	6.454
30.00	-2.22	-0.63	0.00	-4.32	0.00	4.32	356.14	112.94	89.96	81.21	21.07	-0.78	5.943
30.00	-2.22	-0.63	0.00	-4.32	0.00	4.32	300.67	90.20	57.38	53.39	21.07	-0.78	8.831
30.60	-2.11	-0.61	0.00	-3.94	0.00	3.94	300.67	90.20	57.38	53.39	21.89	-0.78	8.083
31.50	-2.00	-0.59	0.00	-3.39	0.00	3.39	300.67	90.20	57.38	53.39	23.13	-0.80	7.009
32.20	-1.84	-0.53	0.00	-2.97	0.00	2.97	300.67	90.20	57.38	53.39	24.11	-0.81	6.181
33.00	-1.74	-0.51	0.00	-2.54	0.00	2.54	300.67	90.20	57.38	53.39	25.24	-0.82	5.349
34.50	-1.55	-0.46	0.00	-1.79	0.00	1.79	300.67	90.20	57.38	53.39	27.40	-0.83	3.866
36.00	-1.37	-0.43	0.00	-1.09	0.00	1.09	300.67	90.20	57.38	53.39	29.60	-0.84	2.501
36.10	-1.35	-0.42	0.00	-1.05	0.00	1.05	300.67	90.20	57.38	53.39	29.74	-0.84	2.415
36.10	-1.35	-0.42	0.00	-1.05	0.00	1.05	325.80	97.74	62.12	58.52	29.74	-0.84	2.208
37.20	-0.77	-0.20	0.00	-0.58	0.00	0.58	325.80	97.74	62.12	58.52	31.37	-0.85	1.236
37.50	-0.74	-0.19	0.00	-0.52	0.00	0.52	325.80	97.74	62.12	58.52	31.81	-0.85	1.119
39.00	-0.54	-0.15	0.00	-0.24	0.00	0.24	325.80	97.74	62.12	58.52	34.04	-0.85	0.580
40.00	-0.32	-0.06	0.00	-0.09	0.00	0.09	325.80	97.74	62.12	58.52	35.52	-0.85	0.255
40.50	-0.26	-0.04	0.00	-0.06	0.00	0.06	325.80	97.74	62.12	58.52	36.27	-0.85	0.184
42.00	-0.03	0.00	0.00	0.00	0.00	0.00	325.80	97.74	62.12	58.52	38.49	-0.85	0.009

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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<u>Load Case:</u> 1.0D + 1.0W							96.56 km/hrServiceability				26 Iterations			
Gust Response Factor :1.10							Wind Importance Factor 0.00							
Dead Load Factor :1.00														
Wind Load Factor :1.00														
42.20	0.00	0.00	0.00	0.00	0.00	0.00	325.80	97.74	62.12	58.52	38.79	-0.85	0.000	

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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Load Case: 0.9D + 1.0W

151.8 km/hr with No Ice (Reduced DL)

26 Iterations

Gust Response Factor :1.10

Wind Importance Factor 0.00

Dead Load Factor :0.90

Wind Load Factor :1.00

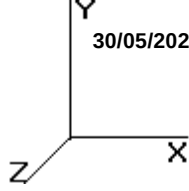
Applied Segment Forces Summary

Seg Elev (m)	Lateral FX (-) (ton)	Axial FY (-) (ton)	Torsion MY (ton- m)	Moment MZ (ton- m)
0.00	0.03	0.00	0.00	0.00
1.50	0.11	0.39	0.00	0.00
3.00	0.11	0.39	0.00	0.00
4.50	0.11	0.39	0.00	0.00
6.00	0.08	0.39	0.00	0.00
6.10	0.04	0.03	0.00	0.00
7.50	0.10	0.32	0.00	0.00
9.00	0.11	0.34	0.00	0.00
10.50	0.11	0.34	0.00	0.00
11.70	0.07	0.28	0.00	0.00
12.00	0.05	0.06	0.00	0.00
13.50	0.11	0.29	0.00	0.00
15.00	0.11	0.29	0.00	0.00
16.50	0.11	0.29	0.00	0.00
17.80	0.08	0.25	0.00	0.00
18.00	0.04	0.03	0.00	0.00
19.50	0.12	0.22	0.00	0.00
21.00	0.12	0.22	0.00	0.00
22.50	0.12	0.22	0.00	0.00
23.90	0.09	0.20	0.00	0.00
24.00	0.04	0.01	0.00	0.00
25.50	0.13	0.22	0.00	0.00
27.00	0.13	0.22	0.00	0.00
28.50	0.10	0.22	0.00	0.00
28.60	0.03	0.04	0.00	0.00
29.00	0.34	0.27	0.00	0.00
29.40	0.07	0.11	0.00	0.00
30.00	0.05	0.09	0.00	0.00
30.60	0.05	0.09	0.00	0.00
31.50	0.05	0.10	0.00	0.00
32.20	0.14	0.14	0.00	0.00
33.00	0.07	0.09	0.00	0.00
34.50	0.10	0.17	0.00	0.00
36.00	0.07	0.17	0.00	0.00
36.10	0.03	0.01	0.00	0.00
37.20	0.52	0.52	0.00	0.00
37.50	0.04	0.04	0.00	0.00
39.00	0.08	0.18	0.00	0.00
40.00	0.22	0.20	0.00	0.00
40.50	0.05	0.06	0.00	0.00
42.00	0.08	0.21	0.00	0.01
42.20	0.01	0.02	0.00	0.00
Totals:	4.20	8.10	0.00	0.01

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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Load Case: 0.9D + 1.0W

151.8 km/hr with No Ice (Reduced DL)

26 Iterations

Gust Response Factor :1.10
Dead Load Factor :0.90
Wind Load Factor :1.00

Wind Importance Factor 0.00

Calculated Forces

Seg Elev (m)	Pu FY (-) (ton)	Vu FX (-) (ton)	Tu MY (ton-m)	Mu MZ (ton-m)	Mu MZ (ton-m)	Resultant Moment (ton-m)	phi Pn (ton)	phi Vn (ton)	phi Tn (ton-m)	phi Mn (ton-m)	Total Deflect (cm)	Rotation (deg)	Ratio
0.00	-8.10	-4.17	0.00	-104.99	0.00	104.99	697.39	221.22	246.74	222.72	0.00	0.00	48.338
1.50	-7.71	-4.08	0.00	-98.74	0.00	98.74	697.39	221.22	246.74	222.72	0.13	-0.09	45.471
3.00	-7.31	-3.98	0.00	-92.62	0.00	92.62	697.39	221.22	246.74	222.72	0.50	-0.18	42.668
4.50	-6.92	-3.88	0.00	-86.65	0.00	86.65	697.39	221.22	246.74	222.72	1.09	-0.27	39.931
6.00	-6.54	-3.80	0.00	-80.84	0.00	80.84	697.39	221.22	246.74	222.72	1.89	-0.35	37.263
6.10	-6.51	-3.77	0.00	-80.46	0.00	80.46	697.39	221.22	246.74	222.72	1.95	-0.35	37.088
6.10	-6.51	-3.77	0.00	-80.46	0.00	80.46	648.59	194.58	185.56	171.63	1.95	-0.35	47.920
7.50	-6.18	-3.67	0.00	-75.18	0.00	75.18	648.59	194.58	185.56	171.63	2.90	-0.42	44.794
9.00	-5.83	-3.57	0.00	-69.68	0.00	69.68	648.59	194.58	185.56	171.63	4.13	-0.52	41.529
10.50	-5.48	-3.46	0.00	-64.33	0.00	64.33	648.59	194.58	185.56	171.63	5.63	-0.62	38.357
11.70	-5.20	-3.39	0.00	-60.17	0.00	60.17	648.59	194.58	185.56	171.63	7.01	-0.69	35.893
11.70	-5.20	-3.39	0.00	-60.17	0.00	60.17	527.32	158.20	125.54	118.98	7.01	-0.69	51.610
12.00	-5.14	-3.35	0.00	-59.16	0.00	59.16	527.32	158.20	125.54	118.98	7.37	-0.71	50.742
13.50	-4.85	-3.25	0.00	-54.13	0.00	54.13	527.32	158.20	125.54	118.98	9.42	-0.85	46.460
15.00	-4.55	-3.14	0.00	-49.26	0.00	49.26	527.32	158.20	125.54	118.98	11.84	-0.99	42.308
16.50	-4.26	-3.03	0.00	-44.55	0.00	44.55	527.32	158.20	125.54	118.98	14.59	-1.11	38.291
17.80	-4.01	-2.95	0.00	-40.61	0.00	40.61	527.32	158.20	125.54	118.98	17.22	-1.20	34.930
17.80	-4.01	-2.95	0.00	-40.61	0.00	40.61	356.14	112.94	89.96	81.21	17.22	-1.20	51.201
18.00	-3.98	-2.91	0.00	-40.02	0.00	40.02	356.14	112.94	89.96	81.21	17.64	-1.22	50.463
19.50	-3.76	-2.80	0.00	-35.65	0.00	35.65	356.14	112.94	89.96	81.21	21.01	-1.35	45.015
21.00	-3.54	-2.68	0.00	-31.46	0.00	31.46	356.14	112.94	89.96	81.21	24.71	-1.47	39.782
22.50	-3.32	-2.56	0.00	-27.44	0.00	27.44	356.14	112.94	89.96	81.21	28.71	-1.58	34.770
23.90	-3.11	-2.47	0.00	-23.86	0.00	23.86	356.14	112.94	89.96	81.21	32.67	-1.66	30.303
23.90	-3.11	-2.47	0.00	-23.86	0.00	23.86	356.14	112.94	89.96	81.21	32.67	-1.66	30.303
24.00	-3.10	-2.43	0.00	-23.61	0.00	23.61	356.14	112.94	89.96	81.21	32.96	-1.67	29.993
25.50	-2.88	-2.30	0.00	-19.97	0.00	19.97	356.14	112.94	89.96	81.21	37.44	-1.75	25.443
27.00	-2.66	-2.17	0.00	-16.52	0.00	16.52	356.14	112.94	89.96	81.21	42.10	-1.81	21.131
28.50	-2.45	-2.07	0.00	-13.27	0.00	13.27	356.14	112.94	89.96	81.21	46.92	-1.87	17.065
28.60	-2.41	-2.03	0.00	-13.07	0.00	13.07	356.14	112.94	89.96	81.21	47.24	-1.87	16.798
29.00	-2.15	-1.69	0.00	-12.25	0.00	12.25	356.14	112.94	89.96	81.21	48.55	-1.88	15.713
29.40	-2.04	-1.61	0.00	-11.58	0.00	11.58	356.14	112.94	89.96	81.21	49.87	-1.89	14.849
30.00	-1.95	-1.56	0.00	-10.61	0.00	10.61	356.14	112.94	89.96	81.21	51.86	-1.91	13.634
30.00	-1.95	-1.56	0.00	-10.61	0.00	10.61	300.67	90.20	57.38	53.39	51.86	-1.91	20.555
30.60	-1.86	-1.51	0.00	-9.68	0.00	9.68	300.67	90.20	57.38	53.39	53.86	-1.92	18.770
31.50	-1.76	-1.45	0.00	-8.32	0.00	8.32	300.67	90.20	57.38	53.39	56.91	-1.96	16.189
32.20	-1.62	-1.31	0.00	-7.30	0.00	7.30	300.67	90.20	57.38	53.39	59.33	-1.99	14.231
33.00	-1.53	-1.24	0.00	-6.25	0.00	6.25	300.67	90.20	57.38	53.39	62.12	-2.01	12.237
34.50	-1.36	-1.14	0.00	-4.39	0.00	4.39	300.67	90.20	57.38	53.39	67.43	-2.05	8.690
36.00	-1.20	-1.06	0.00	-2.68	0.00	2.68	300.67	90.20	57.38	53.39	72.83	-2.07	5.432
36.10	-1.19	-1.04	0.00	-2.57	0.00	2.57	300.67	90.20	57.38	53.39	73.19	-2.07	5.228
36.10	-1.19	-1.04	0.00	-2.57	0.00	2.57	325.80	97.74	62.12	58.52	73.19	-2.07	4.773
37.20	-0.68	-0.50	0.00	-1.43	0.00	1.43	325.80	97.74	62.12	58.52	77.19	-2.08	2.661
37.50	-0.65	-0.46	0.00	-1.28	0.00	1.28	325.80	97.74	62.12	58.52	78.28	-2.09	2.393
39.00	-0.47	-0.37	0.00	-0.59	0.00	0.59	325.80	97.74	62.12	58.52	83.75	-2.09	1.163
40.00	-0.29	-0.15	0.00	-0.22	0.00	0.22	325.80	97.74	62.12	58.52	87.40	-2.09	0.470
40.50	-0.23	-0.09	0.00	-0.15	0.00	0.15	325.80	97.74	62.12	58.52	89.23	-2.09	0.327
42.00	-0.02	-0.01	0.00	0.00	0.00	0.00	325.80	97.74	62.12	58.52	94.71	-2.10	0.010

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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Load Case: 0.9D + 1.0W

151.8 km/hr with No Ice (Reduced DL)

26 Iterations

Gust Response Factor :1.10

Wind Importance Factor 0.00

Dead Load Factor :0.90

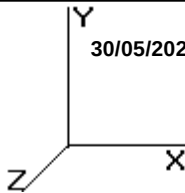
Wind Load Factor :1.00

42.20 0.00 -0.01 0.00 0.00 0.00 0.00 0.00 325.80 97.74 62.12 58.52 95.45 -2.10 0.000

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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Load Case: 1.2D + 1.0W

151.8 km/hr with No Ice

27 Iterations

Gust Response Factor :1.10

Dead Load Factor :1.20

Wind Load Factor :1.00

Wind Importance Factor 0.00

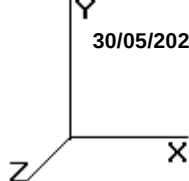
Applied Segment Forces Summary

Seg Elev (m)	Lateral FX (-) (ton)	Axial FY (-) (ton)	Torsion MY (ton- m)	Moment MZ (ton- m)
0.00	0.03	0.00	0.00	0.00
1.50	0.11	0.51	0.00	0.00
3.00	0.11	0.51	0.00	0.00
4.50	0.11	0.51	0.00	0.00
6.00	0.08	0.51	0.00	0.00
6.10	0.04	0.03	0.00	0.00
7.50	0.10	0.43	0.00	0.00
9.00	0.11	0.46	0.00	0.00
10.50	0.11	0.46	0.00	0.00
11.70	0.07	0.37	0.00	0.00
12.00	0.05	0.08	0.00	0.00
13.50	0.11	0.38	0.00	0.00
15.00	0.11	0.38	0.00	0.00
16.50	0.11	0.38	0.00	0.00
17.80	0.08	0.33	0.00	0.00
18.00	0.04	0.04	0.00	0.00
19.50	0.12	0.29	0.00	0.00
21.00	0.12	0.29	0.00	0.00
22.50	0.12	0.29	0.00	0.00
23.90	0.09	0.27	0.00	0.00
24.00	0.04	0.02	0.00	0.00
25.50	0.13	0.29	0.00	0.00
27.00	0.13	0.29	0.00	0.00
28.50	0.10	0.29	0.00	0.00
28.60	0.03	0.06	0.00	0.00
29.00	0.34	0.36	0.00	0.00
29.40	0.07	0.15	0.00	0.00
30.00	0.05	0.12	0.00	0.00
30.60	0.05	0.13	0.00	0.00
31.50	0.05	0.14	0.00	0.00
32.20	0.14	0.19	0.00	0.00
33.00	0.07	0.12	0.00	0.00
34.50	0.10	0.23	0.00	0.00
36.00	0.07	0.23	0.00	0.00
36.10	0.03	0.02	0.00	0.00
37.20	0.52	0.70	0.00	0.00
37.50	0.04	0.05	0.00	0.00
39.00	0.08	0.24	0.00	0.00
40.00	0.22	0.26	0.00	0.00
40.50	0.05	0.08	0.00	0.00
42.00	0.08	0.28	0.00	0.01
42.20	0.01	0.03	0.00	0.00
Totals:	4.20	10.80	0.00	0.01

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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Load Case: 1.2D + 1.0W

151.8 km/hr with No Ice

27 Iterations

Gust Response Factor :1.10
Dead Load Factor :1.20
Wind Load Factor :1.00

Wind Importance Factor 0.00

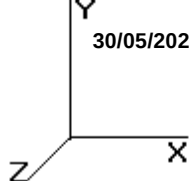
Calculated Forces

Seg Elev (m)	Pu FY (-) (ton)	Vu FX (-) (ton)	Tu MY (ton-m)	Mu MZ (ton-m)	Mu MZ (ton-m)	Resultant Moment (ton-m)	phi Pn (ton)	phi Vn (ton)	phi Tn (ton-m)	phi Mn (ton-m)	Total Deflect (cm)	Rotation (deg)	Ratio
0.00	-10.80	-4.17	0.00	-105.88	0.00	105.88	697.39	221.22	246.74	222.72	0.00	0.00	49.125
1.50	-10.28	-4.08	0.00	-99.62	0.00	99.62	697.39	221.22	246.74	222.72	0.13	-0.10	46.238
3.00	-9.76	-3.99	0.00	-93.50	0.00	93.50	697.39	221.22	246.74	222.72	0.50	-0.19	43.412
4.50	-9.24	-3.89	0.00	-87.52	0.00	87.52	697.39	221.22	246.74	222.72	1.10	-0.27	40.649
6.00	-8.72	-3.81	0.00	-81.68	0.00	81.68	697.39	221.22	246.74	222.72	1.91	-0.35	37.954
6.10	-8.68	-3.78	0.00	-81.30	0.00	81.30	697.39	221.22	246.74	222.72	1.97	-0.35	37.777
6.10	-8.68	-3.78	0.00	-81.30	0.00	81.30	648.59	194.58	185.56	171.63	1.97	-0.35	48.745
7.50	-8.25	-3.69	0.00	-76.00	0.00	76.00	648.59	194.58	185.56	171.63	2.92	-0.42	45.590
9.00	-7.78	-3.59	0.00	-70.47	0.00	70.47	648.59	194.58	185.56	171.63	4.17	-0.53	42.291
10.50	-7.32	-3.48	0.00	-65.09	0.00	65.09	648.59	194.58	185.56	171.63	5.68	-0.63	39.083
11.70	-6.95	-3.41	0.00	-60.91	0.00	60.91	648.59	194.58	185.56	171.63	7.07	-0.70	36.589
11.70	-6.95	-3.41	0.00	-60.91	0.00	60.91	527.32	158.20	125.54	118.98	7.07	-0.70	52.557
12.00	-6.87	-3.38	0.00	-59.88	0.00	59.88	527.32	158.20	125.54	118.98	7.44	-0.72	51.680
13.50	-6.48	-3.28	0.00	-54.82	0.00	54.82	527.32	158.20	125.54	118.98	9.52	-0.86	47.348
15.00	-6.09	-3.17	0.00	-49.91	0.00	49.91	527.32	158.20	125.54	118.98	11.96	-1.00	43.143
16.50	-5.70	-3.06	0.00	-45.15	0.00	45.15	527.32	158.20	125.54	118.98	14.74	-1.12	39.070
17.80	-5.37	-2.98	0.00	-41.17	0.00	41.17	527.32	158.20	125.54	118.98	17.40	-1.22	35.659
17.80	-5.37	-2.98	0.00	-41.17	0.00	41.17	356.14	112.94	89.96	81.21	17.40	-1.22	52.274
18.00	-5.32	-2.95	0.00	-40.58	0.00	40.58	356.14	112.94	89.96	81.21	17.82	-1.23	51.526
19.50	-5.03	-2.83	0.00	-36.16	0.00	36.16	356.14	112.94	89.96	81.21	21.23	-1.37	45.999
21.00	-4.74	-2.71	0.00	-31.91	0.00	31.91	356.14	112.94	89.96	81.21	24.98	-1.49	40.683
22.50	-4.44	-2.59	0.00	-27.85	0.00	27.85	356.14	112.94	89.96	81.21	29.02	-1.60	35.588
23.90	-4.17	-2.50	0.00	-24.22	0.00	24.22	356.14	112.94	89.96	81.21	33.04	-1.68	31.043
23.90	-4.17	-2.50	0.00	-24.22	0.00	24.22	356.14	112.94	89.96	81.21	33.04	-1.68	31.043
24.00	-4.15	-2.46	0.00	-23.97	0.00	23.97	356.14	112.94	89.96	81.21	33.33	-1.69	30.728
25.50	-3.86	-2.33	0.00	-20.28	0.00	20.28	356.14	112.94	89.96	81.21	37.86	-1.77	26.095
27.00	-3.57	-2.20	0.00	-16.78	0.00	16.78	356.14	112.94	89.96	81.21	42.58	-1.83	21.704
28.50	-3.28	-2.10	0.00	-13.48	0.00	13.48	356.14	112.94	89.96	81.21	47.46	-1.89	17.559
28.60	-3.23	-2.06	0.00	-13.27	0.00	13.27	356.14	112.94	89.96	81.21	47.79	-1.89	17.285
29.00	-2.88	-1.71	0.00	-12.45	0.00	12.45	356.14	112.94	89.96	81.21	49.11	-1.90	16.160
29.40	-2.73	-1.63	0.00	-11.76	0.00	11.76	356.14	112.94	89.96	81.21	50.45	-1.92	15.273
30.00	-2.62	-1.58	0.00	-10.78	0.00	10.78	356.14	112.94	89.96	81.21	52.46	-1.93	14.031
30.00	-2.62	-1.58	0.00	-10.78	0.00	10.78	300.67	90.20	57.38	53.39	52.46	-1.93	21.097
30.60	-2.49	-1.53	0.00	-9.83	0.00	9.83	300.67	90.20	57.38	53.39	54.49	-1.95	19.275
31.50	-2.36	-1.48	0.00	-8.45	0.00	8.45	300.67	90.20	57.38	53.39	57.58	-1.99	16.643
32.20	-2.17	-1.33	0.00	-7.42	0.00	7.42	300.67	90.20	57.38	53.39	60.03	-2.01	14.642
33.00	-2.06	-1.26	0.00	-6.36	0.00	6.36	300.67	90.20	57.38	53.39	62.85	-2.04	12.607
34.50	-1.83	-1.16	0.00	-4.46	0.00	4.46	300.67	90.20	57.38	53.39	68.24	-2.07	8.985
36.00	-1.61	-1.08	0.00	-2.73	0.00	2.73	300.67	90.20	57.38	53.39	73.71	-2.10	5.658
36.10	-1.59	-1.05	0.00	-2.62	0.00	2.62	300.67	90.20	57.38	53.39	74.07	-2.10	5.450
36.10	-1.59	-1.05	0.00	-2.62	0.00	2.62	325.80	97.74	62.12	58.52	74.07	-2.10	4.977
37.20	-0.91	-0.51	0.00	-1.46	0.00	1.46	325.80	97.74	62.12	58.52	78.12	-2.11	2.782
37.50	-0.87	-0.47	0.00	-1.31	0.00	1.31	325.80	97.74	62.12	58.52	79.22	-2.11	2.505
39.00	-0.64	-0.38	0.00	-0.61	0.00	0.61	325.80	97.74	62.12	58.52	84.77	-2.12	1.235
40.00	-0.38	-0.15	0.00	-0.23	0.00	0.23	325.80	97.74	62.12	58.52	88.47	-2.12	0.511
40.50	-0.31	-0.10	0.00	-0.15	0.00	0.15	325.80	97.74	62.12	58.52	90.32	-2.12	0.359
42.00	-0.03	-0.01	0.00	0.00	0.00	0.00	325.80	97.74	62.12	58.52	95.87	-2.12	0.013

Pole : 179027
Location : Capulhuac
Height : 42.20 (m)
Base Dia : 106.7 (cm)
Top Dia : 61.00 (cm)
Shape : Round
Taper : 0.000000 (cm/m)

Code: ANSI/TIA-222 Rev H
Risk Category : II
Exposure Category : C
Topographic Category : Method 2 Flat
Base Elev : 0.500 (m)

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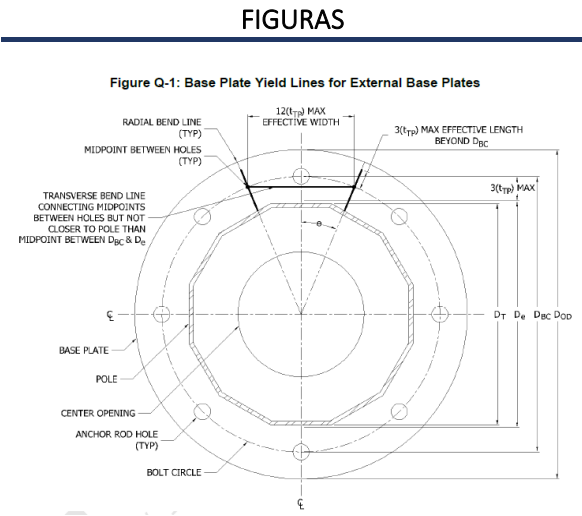
<u>Load Case:</u> 1.2D + 1.0W							151.8 km/hr with No Ice				27 Iterations			
Gust Response Factor :1.10							Wind Importance Factor 0.00							
Dead Load Factor :1.20														
Wind Load Factor :1.00														
42.20	0.00	-0.01	0.00	0.00	0.00	0.00	325.80	97.74	62.12	58.52	96.62	-2.12	0.000	

Analysis Summary

Load Case	Reactions						Max Usage	
	Shear FX (ton)	Shear FZ (ton)	Axial FY (ton)	Moment MX (ton- m)	Moment MY (ton- m)	Moment MZ (ton- m)	Elev (m)	Interactio Ratio
1.2D + 1.0W	4.17	0.00	10.80	0.00	0.00	105.88	11.70	52.56
0.9D + 1.0W	4.17	0.00	8.10	0.00	0.00	104.99	11.70	51.61
1.0D + 1.0W	1.69	0.00	9.00	0.00	0.00	42.61	11.70	21.66

 AMERICAN TOWER®	REVISIÓN DE ANCLAS			Country	México
	ANSI TIA-H Sección Q			Asset Number	179027
	NORMA ANSI TIA H - CHAPTER Q			Asset Name	Capulhuacán
	DESING OF CONECTIONS ON MONOPOLE			Engineer	Misael Sanchez
	Anclas en monopolos			Date	30/05/2024
				Request	CFE-TEIT_SIT-12706003

Datos de Placa base			Reacciones de análisis		
D1=	150	cm	Mu=	105.88	ton-m
D2=	128.40	cm	Ru=	10.8	ton
Espesor PB	1.25	in	Vu=	4.17	ton
Cartabones	Sí				
Cantidad cartabones	22.00				
Espesor cartabones	1/2	in			
Altura cartabones	22.00	cm			
Acero	A-36				
fu=	4077.81	kg/cm2			
fy=	2531	kg/cm3			



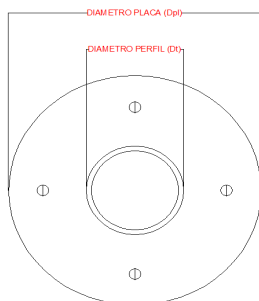
Anclas de placa base de monopolo					
L. Desarrollo=	Desconocido				
nc=	1.27				
DBC=	128.4	cm			
n=	22	No de anclas			
Calidad de acero=	A-36				
fu=	4077.81	kg/cm2	Hilos por in=	6	
fy=	2531	kg/cm2			
Anclas db=	1.5	in			
Ag=	11.40	cm2			
An=	9.07	cm2			
Pu=	15.446	ton			
I.a.r.=	3.27	cm			
Condición:	Caso I				
Put=	15.4457	ton	Puc=	15.4457	ton
Rnt=	36.9698	ton	Rnc=	22.9463	ton
Vu=	0.1895	ton	Vu=	0.1895	ton
Rnv=	23.2454	ton	Rnvc=	6.8839	ton

Formula

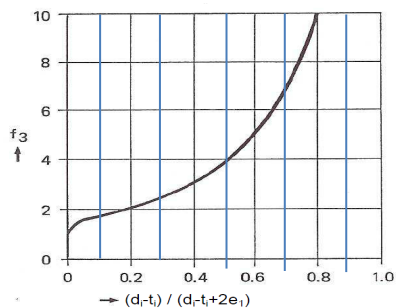
$$I_{ar} \leq 1(d)$$
$$\left[\frac{P_{ut}}{\phi_t R_{nt}} \right]^2 + \left[\frac{V_u}{\phi_v R_{nv}} \right]^2 \leq 1.0$$
$$\left| \frac{P_{uc}}{\phi_c R_{nc}} \right| + \left[\frac{V_u}{\phi_c R_{nvc}} \right]^2 \leq 1.0$$

Anclas					Tensión
CUMPLE	31.04%	<=	105%		
CUMPLE	67.39%	<=	105%		Compresión
Placa Base					
CUMPLE	16.02%	<=	105%		

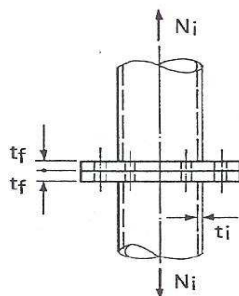
DISEÑO DE BRIDAS ATORNILLADAS TR-02



$T_u = 67,941$ Kg
 $t_f = 2.54$ cm
 $t_i = 1$ cm
 $D_{pl} = 137.19$ cm
 $D_i = 91.4$ cm
 $f_3 = 10$
 tornill 1 cm
 Acero **A-325**
 $f_y = 6475$ kg/cm²
 $N = 16$ tornillos



0.80

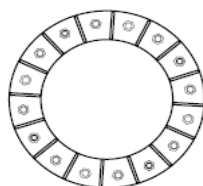
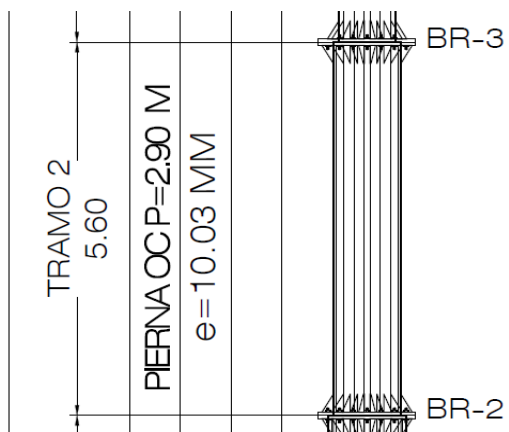


Calculo de espesor de brida

$t_f = 1.37$ cm **PASA**

Numero de tornillos

$N = 6$ tornillos **N° Adecuado**



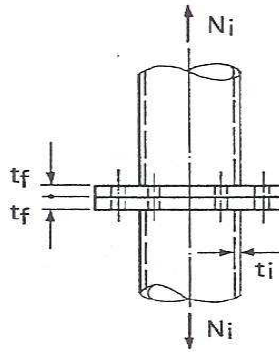
BR-2
 $P = 431$ CM
 $e = 26$ MM
16 TOR Ø24.9 MM



CART
23x20 CM
 $e = 12.8$ MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	67,941	kg
Vu=	3,780	kg
Tornillo	1	in
Area	5.07	cm ²
Acero	A-325	
tb=	20,897	
Kc=	0.82	
n=	16	
tf=	2.54	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	1 in
Area	5.07 cm ²
n=	16

Fn= 385 ton 17.7% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

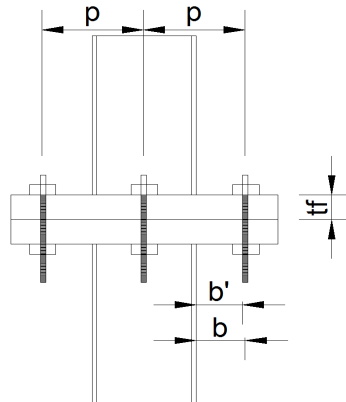
No se realiza esta verificacion

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	47 kg/cm ²
ft=	838 kg/cm ²

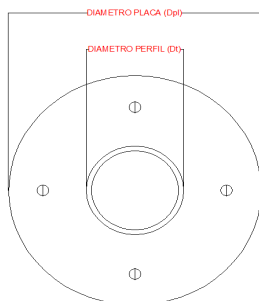
F'nt= 4,745 kg/cm² 17.7% PASA

ACCIÓN SEPARADORA (LRFD)

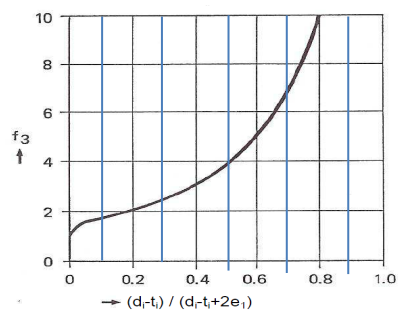
ACERO	A-325
T=	4,246 kg
b=	11.4475 cm
p=	57.1475 cm
Fu=	8440 kg/cm ²
t=	0.63 cm PASA



DISEÑO DE BRIDAS ATORNILLADAS TR-03



Tu= 57,177 Kg
 tf= 2.54 cm
 ti= 0.98 cm
 Dpl 121.27 cm
 Di 76.2 cm
 f3= 10
 tornill 1 cm
 Acero A-325
 fy= 6475 kg/cm²
 N= 16 tornillos



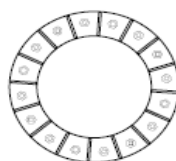
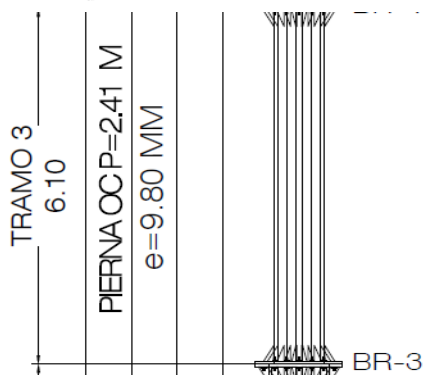
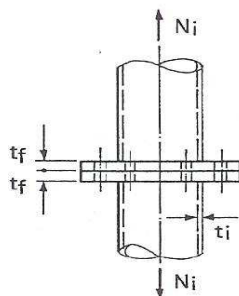
0.77

Calculo de espesor de brida

tf= 1.26 cm PASA

Numero de tornillos

N= 5 tornillos N° Adequado



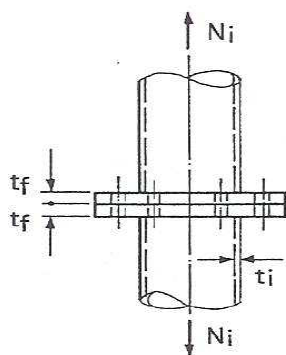
BR-3
 P=381 CM
 e=25.7 MM
 16 TOR Ø24.2 MM



CART
 23x20 CM
 e=13.7 MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	57,177	kg
Vu=	3,410	kg
Tornillo	1	in
Area	5.07	cm ²
Acero	A-325	
tb=	20,897	
Kc=	0.85	
n=	16	
tf=	2.54	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	1 in
Area	5.07 cm ²
n=	16

Fn= 385 ton 14.9% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

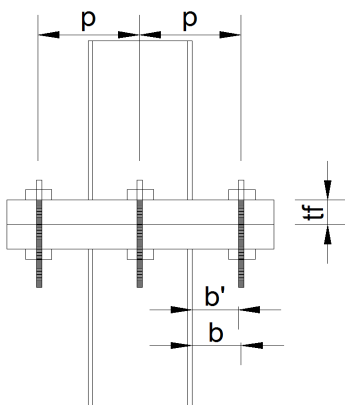
No se realiza esta verificación

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	42 kg/cm ²
ft=	705 kg/cm ²

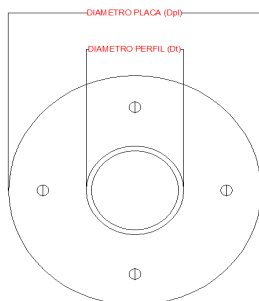
F'nt= 4,745 kg/cm² 14.9% PASA

ACCIÓN SEPARADORA (LRFD)

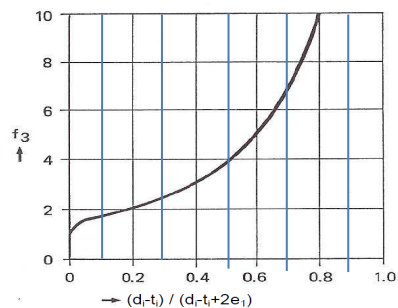
ACERO	A-325
T=	3,574 kg
b=	11.2675 cm
p=	49.3675 cm
Fu=	8440 kg/cm ²
t=	0.62 cm PASA



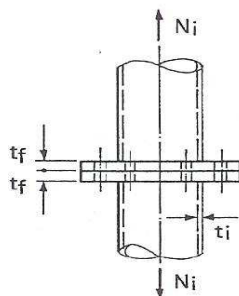
DISEÑO DE BRIDAS ATORNILLADAS TR-04



Tu= 43,865 Kg
 tf= 2.22 cm
 ti= 0.697 cm
 Dpl 106.95 cm
 Di 76.2 cm
 f3= 10
 tornill 7/8 cm
 Acero A-325
 fy= 6475 kg/cm2
 N= 16 tornillos



0.84

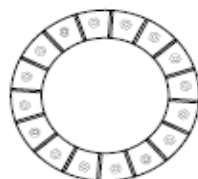
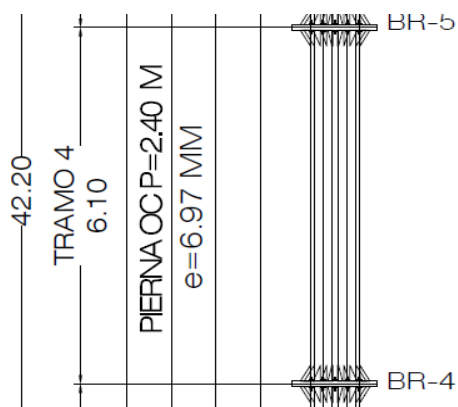


Calculo de espesor de brida

tf= 1.10 cm PASA

Numero de tornillos

N= 5 tornillos N° Adecuado



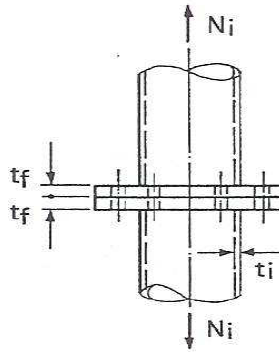
BR-4
 P=336 CM
 e=22.2 MM
 16 TOR Ø22 MM



CART
 16x20 CM
 e=13.8 MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	43,865	kg
Vu=	2,980	kg
Tornillo	7/8	in
Area	3.88	cm ²
Acero	A-325	
tb=	17,941	
Kc=	0.86	
n=	16	
tf=	2.22	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	7/8 in
Area	3.88 cm ²
n=	16

Fn= 295 ton 14.9% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

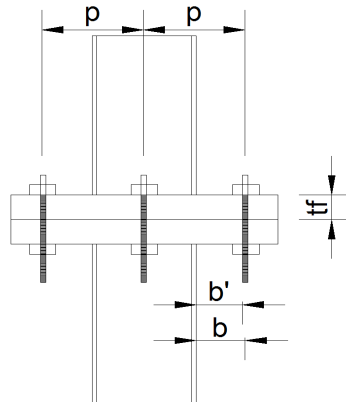
No se realiza esta verificacion

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	48 kg/cm ²
ft=	707 kg/cm ²

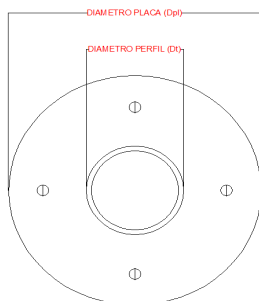
F'nt= 4,745 kg/cm² 14.9% PASA

ACCIÓN SEPARADORA (LRFD)

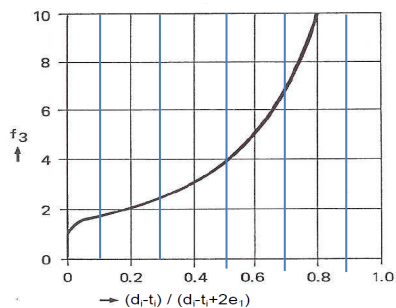
ACERO	A-325
T=	2,742 kg
b=	7.6875 cm
p=	45.7875 cm
Fu=	8440 kg/cm ²
t=	0.46 cm PASA



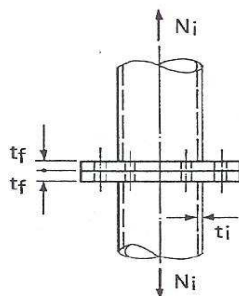
DISEÑO DE BRIDAS ATORNILLADAS TR-05



Tu= 27,158 Kg
 tf= 2.54 cm
 ti= 0.697 cm
 Dpl 105.36 cm
 Di 76.2 cm
 f3= 10
 tornill 7/8 cm
 Acero A-325
 fy= 6475 kg/cm2
 N= 16 tornillos



0.84

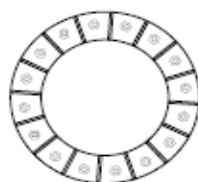
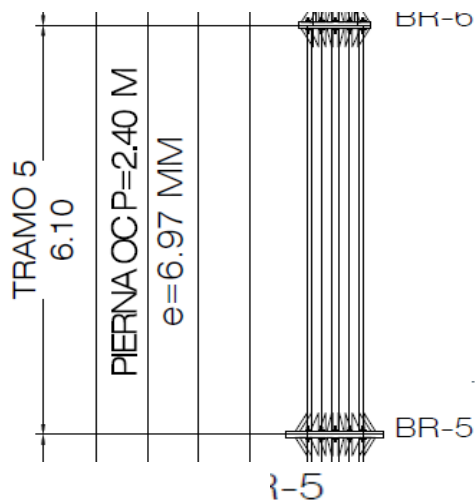


Calculo de espesor de brida

tf= 0.87 cm PASA

Numero de tornillos

N= 4 tornillos N° Adecuado



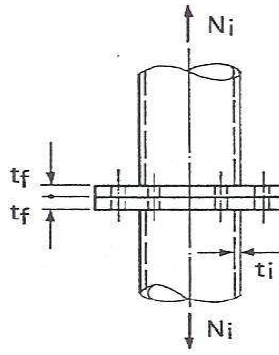
BR-5
 P=331 CM
 e=23.7 MM
 16 TOR Ø21.3 MM



CART
 25x20 CM
 e=13.6 MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	27,158	kg
Vu=	2,500	kg
Tornillo	7/8	in
Area	3.88	cm ²
Acero	A-325	
tb=	17,941	
Kc=	0.92	
n=	16	
tf=	2.54	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	7/8 in
Area	3.88 cm ²
n=	16

Fn= 295 ton 9.2% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

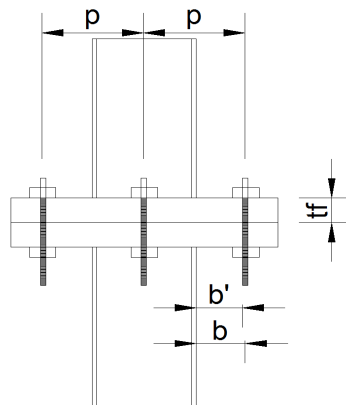
No se realiza esta verificacion

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	40 kg/cm ²
ft=	438 kg/cm ²

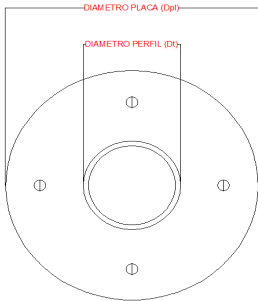
F'nt= 4,745 kg/cm² 9.2% PASA

ACCIÓN SEPARADORA (LRFD)

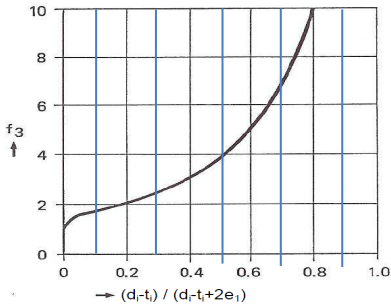
ACERO	A-325
T=	1,697 kg
b=	7.29 cm
p=	45.39 cm
Fu=	8440 kg/cm ²
t=	0.35 cm PASA



DISEÑO DE BRIDAS ATORNILLADAS TR-06



Tu= 12,852 Kg
tf= 1.9 cm
ti= 0.697 cm
Dpl 105.36 cm
Di 61 cm
f3= 10
tornill 3/4 cm
Acero A-325
fy= 6475 kg/cm2
N= 16 tornillos



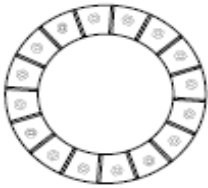
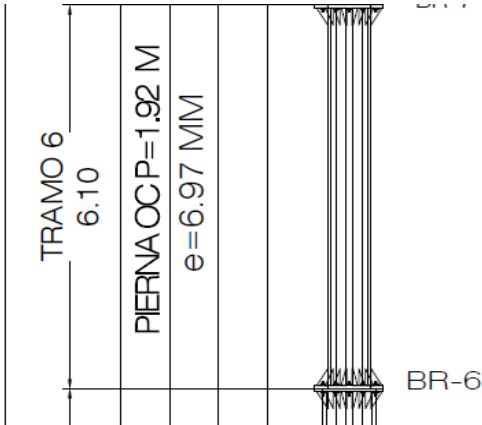
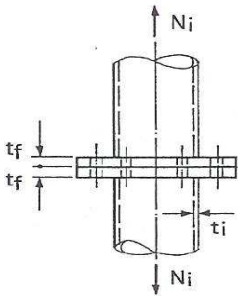
0.74

Calculo de espesor de brida

tf= 0.60 cm PASA

Numero de tornillos

N= 2 tornillos N° Adecuado

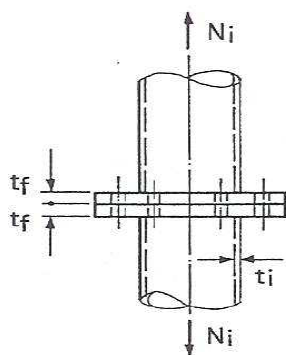


CART
23x20 CM
e=13.0 MM

BR-6
P=331 CM
e=19.5 MM
16 TOR Ø19 MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	12,852	kg
Vu=	1,580	kg
Tornillo	3/4	in
Area	2.85	cm ²
Acero	A-325	
tb=	14,475	
Kc=	0.95	
n=	16	
tf=	1.9	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	3/4 in
Area	2.85 cm ²
n=	16

Fn= 216 ton 5.9% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

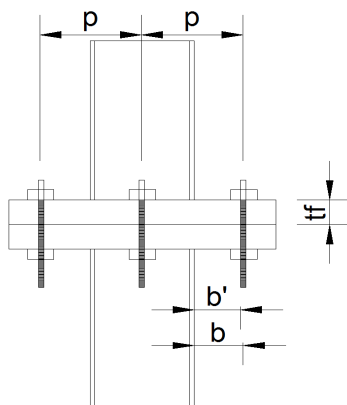
No se realiza esta verificacion

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	35 kg/cm ²
ft=	282 kg/cm ²

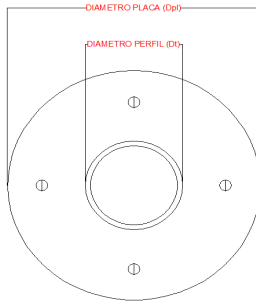
F'nt= 4,745 kg/cm² 5.9% PASA

ACCIÓN SEPARADORA (LRFD)

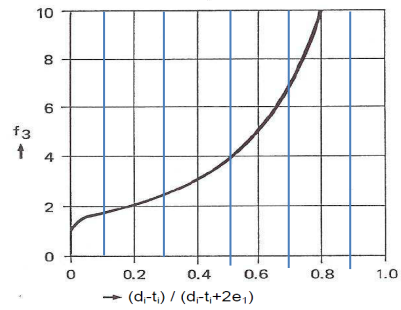
ACERO	A-325
T=	803 kg
b=	11.09 cm
p=	41.59 cm
Fu=	8440 kg/cm ²
t=	0.32 cm PASA



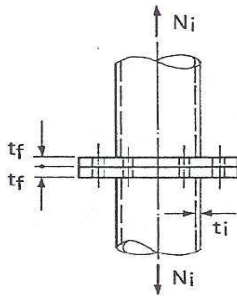
DISEÑO DE BRIDAS ATORNILLADAS TR-07



Tu= 4,116 Kg
 tf= 1.9 cm
 ti= 0.756 cm
 Dpl 109 cm
 Di 61 cm
 f3= 10
 tornill 3/4 cm
 Acero A-325
 fy= 6475 kg/cm2
 N= 16 tornillos



0.72

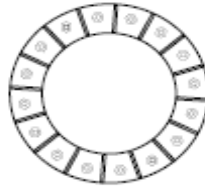
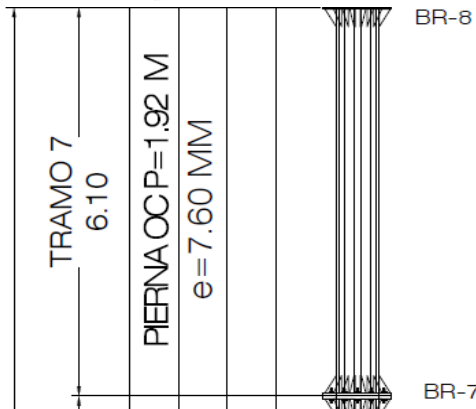


Calculo de espesor de brida

tf= 0.34 cm PASA

Numero de tornillos

N= 1 tornillos N° Adecuado

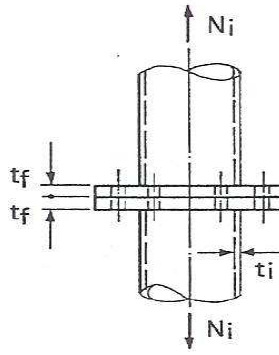


CART
 24x20 CM
 e=13.8 MM

BR-7
 P=345 CM
 e=19.7 MM
 16 TOR Ø18.7 MM

DISEÑO DE BRIDAS ATORNILLADAS

Tu=	4,116	kg
Vu=	1,080	kg
Tornillo	3/4	in
Area	2.85	cm ²
Acero	A-325	
tb=	14,475	
Kc=	0.98	
n=	16	
tf=	1.9	cm



JUNTAS ATORNILLADAS A TENSION (LRFD)

ACERO	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Tornillo	3/4 in
Area	2.85 cm ²
n=	16

Fn= 216 ton 1.9% PASA

TENSION Y CORTANTE COMBINADOS APLASTAMIENTO (LRFD)

No se realiza esta verificacion

Acero	Tornillos A-325A
Fnt=	6,327 kg/cm ²
Fnv=	3,796 kg/cm ²
fv=	24 kg/cm ²
ft=	90 kg/cm ²

F'nt= 4,745 kg/cm² 1.9% PASA

ACCIÓN SEPARADORA (LRFD)

ACERO	A-325
T=	257 kg
b=	12 cm
p=	42.5 cm
Fu=	8440 kg/cm ²
t=	0.19 cm PASA

